Classical Field Theory — Exercise Sheet 10

Residual gauges and charged tails

Solutions

Write $\Box = \partial_t^2 - \Delta$. Use $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$, $\mathbf{B} = \operatorname{curl}\mathbf{A}$ and $\ell = -(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A})$. Lecture 10's theorem assumes a smooth conserved source of compact *spacetime* support and compatible compact smooth potential data. Problems 2–3 explicitly separate a charged background outside that class.

Exercise 10.1 (All residual shifts of Cauchy data). Let a smooth Lorenz-gauge Maxwell potential have initial data $(a_\phi, b_\phi, a_A, b_A)$ as in Lecture 10, with a fixed source ρ. Let χ be the smooth wave with arbitrary real compact smooth data $\chi(0) = p$, $\chi_t(0) = q$. Set $\phi'_{\mathrm{em}} = \phi_{\mathrm{em}} - \chi_t$, $\mathbf{A}' = \mathbf{A} + \nabla\chi$.

(a) Find the increments of all four potential data. Verify directly that both compatibility expressions $b_\phi + \operatorname{div} a_A$ and $-\Delta a_\phi - \operatorname{div} b_A - \rho(0)$ are unchanged. Compare the two initial electromagnetic fields.

(b) Prove that fixing all four potential data rules out every nonzero residual gauge wave in this compact-data class. Contrast this with fixing only the electromagnetic fields. If compactness of the gauge data is dropped, explain why a constant χ is physically trivial.

Solution. The wave equation gives $\chi_{tt}(0) = \Delta p$. Evaluating the potential transformation and its first time derivative gives

$$a'_\phi = a_\phi - q, \quad b'_\phi = b_\phi - \Delta p, \qquad a'_A = a_A + \nabla p, \quad b'_A = b_A + \nabla q.$$

The first compatibility increment is $-\Delta p + \operatorname{div}\nabla p = 0$. The second increment is $\Delta q - \operatorname{div}\nabla q = 0$; the prescribed source is fixed. The electromagnetic data satisfy

$$\mathbf{E}'_0 = -\nabla(a_\phi - q) - (b_A + \nabla q) = \mathbf{E}_0, \qquad \mathbf{B}'_0 = \operatorname{curl}(a_A + \nabla p) = \mathbf{B}_0.$$

The full fields also agree by the gauge identity of Lecture 9, and Lorenz gauge persists because $\ell' = \ell + \Box\chi = 0$.

If all four potential data coincide, the first and third increments imply $q = 0$ and $\nabla p = 0$. A smooth function with zero gradient on connected $\mathbb{R}^3$ is constant; compact support forces that constant to be zero. Lecture 4 then gives the zero wave. By contrast, arbitrary compact p, q produce transformations with identical electromagnetic fields, many of which have nonzero potential increments. Without compactness, a constant χ changes no potential since every component of $\mathrm{d}\chi$ is zero. These distinctions concern uniqueness of potentials versus uniqueness of fields.

Exercise 10.2 (A smooth charge with a Coulomb tail). Choose a radial nonnegative $b \in C_c^\infty(\mathbb{R}^3)$ with $\int b \mathrm{d}^3x = 1$ and support in $\overline{B_R(0)}$, $R > 0$. Fix real $Q \neq 0$ and prescribe the static source $\rho_Q(x) = Qb(x)$, $\mathbf{J} = 0$ for all time. For $r > 0$ define, writing $\rho_Q(s)$ for its radial profile,

$$M_Q(r) = 4\pi \int_0^r s^2 \rho_Q(s) \mathrm{d}s, \quad E_r(r) = \frac{M_Q(r)}{4\pi r^2}, \quad \phi_Q(r) = \int_r^\infty E_r(s) \mathrm{d}s.$$

(a) Prove that $E_Q(x) = E_r(|x|)x/|x|$ and $\phi_Q(|x|)$ extend smoothly across zero. Prove Gauss's law there as well as away from zero. Show that $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q, 0)$ solves Maxwell in Lorenz gauge.

(b) Determine the exact potential and field for $r \geq R$. Prove $\phi_Q \notin L^2$, hence $\phi_Q \notin H^1$, while $\nabla\phi_Q \in L^2$ and the field energy is finite. Calculate the electric energy outside B_R.

(c) Identify the failed hypotheses of Lecture 10's theorem. Why does conservation alone not force $Q = 0$ for this source?

Solution. Changing variables $s = ar$ in M_Q gives, for $x \neq 0$,

$$E_Q(x) = x \int_0^1 a^2 \rho_Q(ax) \mathrm{d}a.$$

The right side is smooth for every x, by differentiating under a compact parameter integral, and is $O(|x|)$ near zero. It supplies the extension. Taking its divergence gives an identity valid also at zero:

$$\operatorname{div} E_Q = \int_0^1 [3a^2 \rho_Q(ax) + a^3 x \cdot \nabla\rho_Q(ax)] \mathrm{d}a = \int_0^1 \partial_a [a^3 \rho_Q(ax)] \mathrm{d}a = \rho_Q(x).$$

The potential integral converges at infinity since M_Q is constant there, and at zero since $E_r(r) = O(r)$. Its radial derivative is $-E_r$. More explicitly, its smooth extension can be written

$$\phi_Q(x) = \phi_Q(0) - \int_0^1 E_Q(\tau x) \cdot x \mathrm{d}\tau.$$

This equals the radial integral for $x \neq 0$, so it is a smooth extension and $\nabla\phi_Q = -E_Q$ there and, by continuity, at zero. The field is static and has zero curl; $\mathbf{A} = 0$ gives zero magnetic field. Gauss's law was proved above, Ampère's law has both sides zero, and the remaining two equations follow from the potential. Its Lorenz function is $-(\partial_t \phi_Q + \operatorname{div} 0) = 0$.

For $r \geq R$, normalization gives $M_Q(r) = Q$. Hence

$$E_r(r) = \frac{Q}{4\pi r^2}, \qquad \phi_Q(r) = \frac{Q}{4\pi r}.$$

The potential's exterior squared norm contains $Q^2 \int_R^\infty \mathrm{d}r/(4\pi) = \infty$. The gradient's exterior squared norm is instead

$$\int_{|x|>R} |\nabla\phi_Q|^2 \mathrm{d}^3x = \frac{Q^2}{4\pi} \int_R^\infty r^{-2} \mathrm{d}r = \frac{Q^2}{4\pi R}.$$

Smoothness on the interior ball makes its remaining gradient integral finite. The electric energy is half that norm squared, so its exterior part is $Q^2/(8\pi R)$ and its total field energy is finite.

The source persists for all time and the scalar potential data are neither compactly supported nor in H^1. Lecture 10's compact-source/data theorem therefore does not apply.

Conservation still makes the total charge constant, but there is no early time at which this static source vanishes. The step that forced neutrality for a compact spacetime source is absent.

Exercise 10.3 (An affine charged evolution class). Keep the background of Exercise 10.2. Let $(\rho_1, J_1) \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R} \times \mathbb{R}^3)$ be conserved. Give compact smooth perturbation data $(a_{1\phi}, b_{1\phi}, a_{1A}, b_{1A})$ satisfying

$$b_{1\phi} + \operatorname{div} a_{1A} = 0, \qquad -\Delta a_{1\phi} - \operatorname{div} b_{1A} = \rho_1(0).$$

Construct a smooth Lorenz-gauge solution for total source $(\rho_Q + \rho_1, J_1)$ and total initial data obtained by adding the static background. Prove its total charge is Q, its electric and magnetic fields have finite energy on each slice, and $\phi_{\mathrm{em}}(t) \in \phi_Q + H^1(\mathbb{R}^3)$ while $\phi_{\mathrm{em}}(t) \notin H^1$. State the uniqueness conclusion actually supplied by Lecture 10. Explain why this construction does not put the charged background in a Schwartz canonical phase space.

Solution. The compact source and the displayed perturbation conditions are exactly the hypotheses of Theorem 10.2. It supplies a smooth global perturbation (ϕ_1, A_1) in Lorenz gauge. Its construction is a retarded compact-source field plus a homogeneous wave with compact corrected data. Lecture 6's support bound and Lecture 5's finite propagation put that perturbation and all its derivatives in a common spatial ball on each bounded time interval. In particular, on each slice $\phi_1 \in H^1$ and its electromagnetic fields are square integrable.

Maxwell's equations and the Lorenz condition are linear. Adding the fixed background gives

$$\phi_{\mathrm{em}} = \phi_Q + \phi_1, \qquad \mathbf{A} = A_1, \qquad \mathbf{E} = E_Q + E_1, \qquad \mathbf{B} = B_1.$$

These have the total source and the prescribed total data. The compact conserved perturbation current has zero total charge by Lecture 10's early-time argument. Thus $\int (\rho_Q + \rho_1) = Q$ at every time. Both electric summands and the magnetic field are in L^2, so the total field energy is finite, including their integrable electric cross term.

Since $\phi_1(t) \in H^1$, the scalar potential lies in $\phi_Q + H^1$. If it also lay in H^1, subtraction of ϕ_1 would put ϕ_Q in H^1, contradicting the preceding problem. Subtracting the fixed smooth background from any other C^2 Lorenz-gauge Maxwell solution with these total potential data gives a perturbation covered by Lecture 10's uniqueness conclusion. Thus the total potential is unique in that class with those full data. Residual transformations with different potential data remain possible as in Problem 1. This is a concrete construction around one fixed background, not a general existence theorem for arbitrary charged tails. Its potential has a $1/r$ tail and is not Schwartz; no Schwartz field bracket on that background has been established by adding the two solutions.

Exercise 10.4 (Local potentials with no global real one-form). On the oriented unit sphere, with area form $\sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$, let $g_{\mathrm{mag}} \neq 0$ be real. On $U_N = S^2 \setminus \{\text{south pole}\}$ and $U_S = S^2 \setminus \{\text{north pole}\}$, respectively, take

$$A_N = g_{\mathrm{mag}}(1 - \cos\theta)\mathrm{d}\varphi, \qquad A_S = -g_{\mathrm{mag}}(1 + \cos\theta)\mathrm{d}\varphi.$$

These are spatial real one-form potentials; the task uses exterior calculus from *Topology and Geometry*, Part I, not bundle theory as a prerequisite.

(a) Verify smoothness at each included pole, compute the common curvature $F = \mathrm{d}A_N = \mathrm{d}A_S$, and find its total flux.

(b) Find a gauge function on a simply connected azimuthal sector in the overlap. Prove that there is no single-valued real gauge function for this shift on the whole overlap, and no globally smooth real one-form A on S^2 with $\mathrm{d}A = F$.

The interpretation by nontrivial bundles and single-valued charged transition phases, including the allowed flux quantization, is a later topic: *Topology and Geometry*, Lectures 20–22. No bundle construction or quantization theorem is assumed in this exercise.

Solution. Away from the coordinate poles, $\mathrm{d}\varphi = (X\mathrm{d}Y - Y\mathrm{d}X)/(X^2 + Y^2)$ and $X^2 + Y^2 = (1 - Z)(1 + Z)$. Thus

$$A_N = \frac{g_{\mathrm{mag}}}{1 + Z}(X\mathrm{d}Y - Y\mathrm{d}X), \qquad A_S = -\frac{g_{\mathrm{mag}}}{1 - Z}(X\mathrm{d}Y - Y\mathrm{d}X).$$

The first expression is smooth on U_N, including the north pole; the second is smooth on U_S, including the south pole. Their exterior derivatives in the angular region agree and extend to the smooth two-form

$$F = g_{\mathrm{mag}} \sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi, \qquad \int_{S^2} F = 4\pi g_{\mathrm{mag}}.$$

On a sector where φ is single-valued, $A_N - A_S = 2g_{\mathrm{mag}}\mathrm{d}\varphi = \mathrm{d}\chi_{NS}$ with $\chi_{NS} = 2g_{\mathrm{mag}}\varphi$. On the full overlap, integrate the difference around the positively oriented equator. Its integral is $4\pi g_{\mathrm{mag}} \neq 0$. An exact one-form of a single-valued real function has zero integral on a closed curve, so no such global overlap function exists. If a global smooth real A on S^2 had $\mathrm{d}A = F$, Stokes on the closed sphere would give $\int F = 0$, again a contradiction. Local smooth potentials can therefore describe a curvature which has no single global real one-form potential. Bundle transition phases require additional structure and consistency conditions, left to the stated later course.